

# SatQuery-Edge

Offline Satellite Intelligence Investigation Report  
SIH 26167 · ISRO · Prototype · Offline Mode

|                  |                       |
|------------------|-----------------------|
| Investigation ID | SQE-20260902-66JSDG   |
| Timestamp        | 2026-09-02 18:54:26   |
| Intent           | text_guided_grounding |
| Tools Used       | grounding, geospatial |

## Query

mark the water body

## Final Answer

Investigation complete. Evidence Score: 71/100.

## Evidence Score

| Component            | Code | Value      | Weight |
|----------------------|------|------------|--------|
| Model Agreement      | M    | 0.650      | 0.30   |
| Sensor Reliability   | S    | 0.860      | 0.20   |
| Spatial Consistency  | P    | 0.650      | 0.20   |
| Temporal Consistency | T    | 0.820      | 0.15   |
| Input Quality        | I    | 0.609      | 0.15   |
|                      |      |            |        |
| FINAL SCORE          |      | 71.1 / 100 |        |

## Validation

Status: **PASS** ✓

- 1 image(s) detected — count OK.
- Image 1: Quality OK (mean=100, std=42.7).

## Specialist Tool Outputs

**GROUNDING** — SUCCESS — 0 region(s)

Grounding DINO + SAM found no candidate regions matching 'water'.

**GEOSPATIAL** — SUCCESS — 0 region(s)

Geospatial conversion complete.  
Converted 0 region(s) to geographic coordinates.  
Total approximate area: 0.00 km².

■ Approximate demo georeferencing applied.

Demo center: (20.296°N, 85.824°E)  
Coordinates are NOT derived from real GeoTIFF metadata.

### Execution Trace

[18:54:26] QUERY RECEIVED: mark the water body  
[18:54:26] ROUTER DECISION: Intent: text\_guided\_grounding  
Selected tools: grounding, geospatial  
Images required: 1  
[18:54:26] VALIDATION: Status: PASS  
1 image(s) detected – count OK.  
Image 1: Quality OK (mean=100, std=42.7).  
[18:54:26] TOOL EXECUTING: grounding → started  
[18:54:26] TOOL OUTPUT (grounding): Success – 0 region(s) detected  
Grounding DINO + SAM found no candidate regions matching 'water'.  
[18:54:26] TOOL EXECUTING: geospatial → started  
[18:54:26] TOOL OUTPUT (geospatial): Success – 0 region(s) detected  
Geospatial conversion complete.  
Converted 0 region(s) to geographic coordinates.  
Total approximate area: 0.00 km².  
[18:54:26] EVIDENCE FUSION: Score: 71.1/100  
M=0.650  
S=0.860  
P=0.650  
[18:54:26] FINAL SCORE: 71/100  
[18:54:26] ANSWER: Investigation complete. Evidence Score: 71/100.

### Tool Versions

| Component               | Version                    |
|-------------------------|----------------------------|
| SatQuery-Edge Prototype | 1.0.0-SIH26167             |
| Python OpenCV           | 4.11.0                     |
| NumPy                   | 1.26.4                     |
| Streamlit               | 1.62.0                     |
| Report Engine           | ReportLab PDF              |
| Mode                    | OFFLINE — No external APIs |

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. — SIH 26167 · ISRO · SatQuery-Edge